

An explicit counterexample to the Auslander–Reiten conjecture

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Abstract

We disprove the Auslander–Reiten conjecture for Artin algebras. We construct a finite-dimensional algebra Λ over $k = \mathbb{F}_2(q, H_1, H_2)$ and a finite-dimensional nonprojective left module Z such that $\text{Ext}_\Lambda^i(Z, Z) = \text{Ext}_\Lambda^i(Z, \Lambda) = 0$ for every $i > 0$. The module is Gorenstein-projective, so the same example also disproves the Gorenstein-projective conjecture. Both counterexamples persist after every extension of k .

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1 Introduction

The Auslander–Reiten conjecture asks whether a finitely generated module M over an Artin algebra R must be projective when

$$\mathrm{Ext}_R^i(M, M \oplus R) = 0 \quad \text{for every } i > 0. \quad (1.1)$$

Here an Artin algebra is finite over a commutative Artinian central base ring. In particular, every finite-dimensional algebra over a field is an Artin algebra. The conjecture originates in the work of Maurice Auslander and Idun Reiten on the generalized Nakayama conjecture [2]; its module and generator formulations are recalled in [12, Introduction].

We give a counterexample. A module is called *Gorenstein-projective* if it is a cokernel in a doubly infinite exact complex of finite projectives that remains exact after applying $\mathrm{Hom}_R(-, R)$. The *Gorenstein-projective conjecture* asks whether a finite Gorenstein-projective module M with $\mathrm{Ext}_R^i(M, M) = 0$ for every $i > 0$ must be projective [16, p. 2]. The ambient algebra is not assumed to be Gorenstein. Total acyclicity gives $\mathrm{Ext}_R^i(M, R) = 0$ for $i > 0$, so this conjecture is precisely the restriction of the Auslander–Reiten conjecture to Gorenstein-projective modules.

Theorem 1.1. *Let $k = \mathbb{F}_2(q, H_1, H_2)$, where q, H_1, H_2 are algebraically independent. There exist a finite-dimensional k -algebra Λ and a finite-dimensional nonprojective left Λ -module Z such that*

$$\mathrm{Ext}_\Lambda^i(Z, Z) = 0 = \mathrm{Ext}_\Lambda^i(Z, \Lambda) \quad \text{for every integer } i \geq 1.$$

The module Z is Gorenstein-projective. The semisimple quotient is $\Lambda/\mathrm{rad} \Lambda \simeq k^8$, so the algebra is split with eight isomorphism classes of simple modules; also $\mathrm{rad}(\Lambda)^4 \neq 0$. These conclusions persist after every field extension of k .

Thus both conjectures fail for Artin algebras, including finite-dimensional algebras over an algebraically closed field of characteristic two: one can extend scalars to an algebraic closure of k . The parameters play essential roles in the proof. The powers of q give an infinite resolution with uniform kernel formulas, and H_1/H_2 has infinite multiplicative order. The algebraic closure in this assertion contains these transcendental elements.

The final algebra is triangular and noncommutative. The companion article [17, Theorem 1.1 and Corollary 1.2] constructs a symmetric counterexample over the same field by a different bimodule construction and a transfer requiring control of negative complete self-extensions, and develops the homological consequences of its associated endomorphism algebra. The contribution here is the explicit eight-simple Gorenstein-projective example and its persistence under every field extension. Each paper contains its own complete proof. The commutative restriction is not resolved by either construction; see [5, Appendix A] for its relationship with the various Auslander and Tachikawa conditions. The generator formulation also fails: $Z \oplus \Lambda$ is a nonprojective generator with no positive self-extensions. Indeed Ext is additive in both arguments, and all positive Ext groups with first argument Λ vanish.

1.1 Earlier constructions and restricted results

The positive cases developed along several lines. For the commutative Noetherian formulation, Auslander, Ding and Solberg treated complete local complete intersections by lifting across

regular sequences [1, Proposition 1.9]. Huneke and Leuschke proved the conjecture for excellent commutative Cohen–Macaulay normal domains containing the rational numbers, among other classes [12, Theorem 0.1]. Christensen and Holm established a general cohomological criterion: every left-noetherian ring satisfying Auslander’s condition satisfies the conjecture [5, Theorem A]. That condition requires, for each fixed finitely generated first module, a common bound after which Ext vanishes for every finitely generated second module whose Ext groups vanish eventually.

Algebras with radical cube zero form another important positive class. Huneke, Şega and Vraciu treated commutative Artinian local rings [13, Theorem 4.1(1)]. For split finite-dimensional local algebras with radical cube zero, Ringel and Zhang proved the stronger assertion that every nonprojective module with no positive Ext to the algebra has a nonzero first self-extension [18, Theorem 1.5]. Zhang and Zhou subsequently proved the conjecture for split finite-dimensional algebras with radical cube zero [23, Theorem 1.1]. Their result for Igusa–Todorov Artin algebras includes all Artin algebras with radical cube zero [24, Theorem 3.2 and Corollary 4.4]. The algebra constructed here has nonzero fourth radical power, as proved in Proposition 9.1.

Schulz’s construction of a nonprojective module without self-extensions [19] is an important antecedent, but its particular skew-field example lies outside the Artin-algebra setting above; see [7, Introduction]. A separate distinction is between all-degree and eventual vanishing: both summands in (1.1) must vanish in every positive degree.

Schulz also exhibited two-dimensional modules over quantum exterior algebras whose syzygies move through a parameter family and whose self-extensions vanish eventually, while the first self-extension remains nonzero [20, Example 7]. The corresponding quantum exterior algebra occurs as a corner of our starting algebra. We use the same parameter-shifting mechanism for a resolution over a larger algebra, then remove every positive self-extension by two independent twists. For indecomposable weakly symmetric algebras over an algebraically closed field with $J^3 = 0 \neq J^2$, Erdmann characterizes those admitting nonprojective modules with eventually vanishing self-extensions [9]; eventual vanishing alone does not give (1.1). Xia proves the Auslander–Reiten conjecture for quantum complete intersections [22], including the quantum exterior algebras used in Schulz’s parameter-shifting examples.

Vanishing in a finite range also differs from the hypothesis of (1.1). Böhmeler and Marczink [3, Theorem 3.1 and the end of Section 3] construct a characteristic-two commutative self-injective example with vanishing first and second self-extensions, and nonzero third self-extension.

Tachikawa’s second conjecture is the self-injective case of (1.1). Enomoto proves that this conjecture for the two-fold trivial extension of an Artin algebra implies the Auslander–Reiten conjecture for that algebra [8, Theorem 3.1]. His reduction produces a self-injective algebra. The symmetric construction in our companion and the structural assertions of Theorem 1.1 are separate contributions.

1.2 How the construction removes self-extensions

The proof has two parts. First we give a conversion principle: a suitable stable map $v : X \rightarrow F \otimes_A X$ over a symmetric algebra A produces a module over

$$\Lambda = \begin{pmatrix} A & 0 \\ F & A \end{pmatrix}.$$

Its positive self-extensions are controlled by the difference between the actions of a self-extension of X on the two sides of v . Total acyclicity separately gives the vanishing of Ext into Λ . This principle is stated and proved before the example, so the subsequent construction has an explicit target.

We then build the required A, F, X, v . Write D for the k -linear dual. A ten-dimensional algebra C has a simple module s whose Ext groups are especially sparse. Passing to its symmetric trivial extension $T = C \ltimes DC$ produces $\text{Ext}_T^*(s, s) = k[\tau]$ with $|\tau| = 3$. An explicit Hochschild cocycle realizes τ and supplies a homotopy compatible with twists of T . For $A = T \otimes T$ and $X = s \otimes s$, the nonnegative self-extension algebra is $k[\tau_1, \tau_2]$.

Write $\mathcal{C}$ for the bimodule obtained from the two cones of these classes. In the stable category, maps factoring through projective modules are identified with zero, and the shift $[1]$ is inverse syzygy. The stable maps from X to $(\mathcal{C} \otimes_A X)[a]$ form a line for $a = 0$ and $a = 3$ and vanish in every other integer degree. This profile supplies a common target for two differently twisted maps.

For a nonzero scalar H , let h_H fix C and scale DC by H , and let U_H be A with right action twisted by $h_H \otimes h_H$. Its tensor functor fixes X but multiplies its degree- $3m$ self-extensions by H^{-m} . Using the symmetrizing trace and the cocycle's boundary identity, we construct maps $U_H \rightarrow \mathcal{C}[3]$ whose evaluations on X are nonzero. After rescaling the maps for H_1 and H_2 , their evaluations are the same nonzero element of that degree-three line. Their finite fiber F then has a single diagonal stable map $v : X \rightarrow F \otimes_A X$ in degree zero, with both projections equal to id_X . In positive degrees, projection identifies the stable maps from X to $(F \otimes_A X)[a]$ with two copies of the self-extensions of X . In degree $3m > 0$, the comparison takes two independently chosen self-extension classes g, j to $F(g)v + v[3m]j$, where $F(g)$ denotes tensoring with F . Under these projections its matrix is

$$\begin{pmatrix} H_1^{-m} & 1 \\ H_2^{-m} & 1 \end{pmatrix} \otimes \text{id}_{k[\tau_1, \tau_2]_{3m}}.$$

This matrix is invertible for every $m > 0$. In degree zero, the comparison is surjective with a nonzero kernel. The conversion principle therefore removes all positive self-extensions while keeping a nonzero stable endomorphism, which proves nonprojectivity.

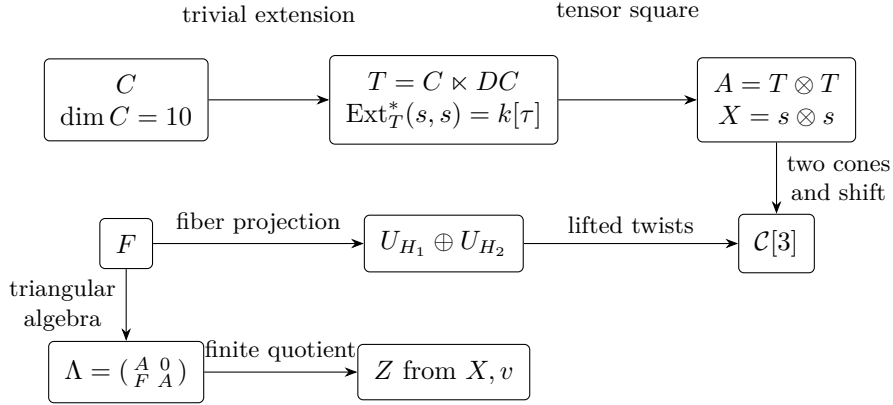


Figure 1: The algebraic and homological stages of the construction. The middle row begins the stable fiber triangle; the other arrows indicate construction steps. The finite bimodule F is the ordinary kernel in (8.3) before Λ is formed. Evaluating this fiber on X gives the diagonal stable map v ; the final module is the finite quotient (3.2), using both X and v .

The main construction-specific ingredients are the explicit cocycle with its twisted boundary identity, the lift through both cones, and the resulting uniform comparison of the two twist actions. The surrounding tools have a substantial history. Trivial extensions and their Hochschild complexes are studied in [6]; Hochschild evaluation as a characteristic action is treated in [15, Section 4.3]. Bimodules attached to Hochschild classes and their iterated tensor products occur in the support-variety constructions of Erdmann, Holloway, Snashall,

Solberg and Taillefer [10, Section 2]. We use the natural symmetric stable duality of [14, Section 2] and the complete-resolution methods of [21]. Gorenstein-projective modules over triangular matrix rings are studied in [11]. We supply the particular comparison and finiteness arguments needed here.

For reference, the objects and their roles are summarized in Table 1. The finite cocycle table appears in Appendix A; its identities are finite algebra calculations. The resolution, duality, cone, and complete-resolution arguments establish the assertions in all degrees.

Section 2 fixes the stable-category conventions, and Section 3 proves the conversion principle. Sections 4 and 5 give the starting algebra and its self-extension calculation. Section 6 computes the two-cone profile; Sections 7 and 8 construct the lifts and the finite fiber. Section 9 proves the structural and scalar-extension assertions.

Object	Ambient algebra	Role or dimension
C	k	ten-dimensional algebra
$T = C \ltimes DC, s$	k, T	symmetric algebra of dimension 20; simple
$A = T \otimes T, X = s \otimes s$	k, A	symmetric algebra of dimension 400; simple
$\mathcal{C}$	stable bimodules over A	two cones of degree-three classes
U_{H_i}	A -bimodules	right twists of A
F	A -bimodules	finite fiber, projective on both sides
$v : X \rightarrow F \otimes_A X$	stable A -modules	both projections equal id_X
Λ, Z	k, Λ	dimension $800 + \dim_k F$; finite module

Table 1: Objects used in the proof. Shifts are homological.

2 Stable maps and complete resolutions

We first fix the language needed to pass from extension classes to an ordinary finite module. All algebras and modules in the paper are finite-dimensional over the indicated field. Modules are left modules unless a side is specified. Write $DV = \text{Hom}_k(V, k)$ and $R^e = R \otimes_k R^{\text{op}}$; an R -bimodule is a left R^e -module. All tensor products without a subscript are over k .

Complexes have homological differentials $d_i : P_i \rightarrow P_{i-1}$ and shifts $P[a]_i = P_{i-a}$. Thus a degree- a cohomology class is represented by a chain map $P \rightarrow Q[a]$. Throughout the construction

$$k = \mathbb{F}_2(q, H_1, H_2),$$

where the three parameters are algebraically independent. Characteristic two removes all differential, shift, and interchange signs.

For any finite algebra R , the space $\underline{\text{Hom}}_R(M, N)$ is the quotient of $\text{Hom}_R(M, N)$ by maps factoring through finite projective modules. An algebra R is *symmetric* if it has a linear form $t : R \rightarrow k$ such that $t(ab) = t(ba)$ and $(a, b) \mapsto t(ab)$ is nondegenerate. The resulting bimodule isomorphism $R \simeq DR$ shows that finite projective R -modules are injective. Every finite module embeds in a finite projective: dualize a finite projective surjection onto its dual.

For symmetric R , set $M[-1] = \Omega M$, the kernel of a projective surjection onto M , and set $M[1]$ to be the cokernel of an embedding of M into a projective. These are well defined on stable maps and inverse to each other there. Lifting through projective surjections and extending into injectives give the maps on shifts; different choices differ by projective factorizations. A short exact sequence induces a stable triangle and long exact sequences of stable Hom groups in all integer degrees. One concrete cone of $u : M \rightarrow N$ is the cokernel of $(u, i) : M \rightarrow N \oplus I$, where $i : M \hookrightarrow I$ is a projective embedding. Its sequence is

$$M \xrightarrow{u} N \longrightarrow \text{cone}(u) \longrightarrow M[1].$$

The usual Hom and Ext sequences prove exactness in positive degrees. At degree zero, maps through a projective lift through a surjection and extend across an injection, giving exactness after quotienting by those maps. Shifting gives every other degree. In particular,

$$\underline{\mathrm{Hom}}_R(M, N[a]) \simeq \mathrm{Ext}_R^a(M, N) \quad (a > 0), \quad (2.1)$$

compatibly with composition and Yoneda products.

A *complete resolution* of M over a symmetric algebra is a doubly infinite exact complex P of finite projectives with $M = \mathrm{coker}(d_1 : P_1 \rightarrow P_0)$. Splicing a projective resolution and an injective coresolution constructs one. A complex of finite projectives over an arbitrary finite algebra is *totally acyclic* if both it and $\mathrm{Hom}_R(P, R)$ are exact. Because every term is finitely generated, exactness into R also gives exactness into every projective target, by direct sums and summands. A module is *Gorenstein-projective* if it is such a cokernel in a totally acyclic complex. Complete resolutions over symmetric algebras are totally acyclic. We will prove the precise comparison with full Hom complexes in the next section; this also applies over the final algebra, which need not be self-injective.

For an A -bimodule F , the notation FM means $F \otimes_A M$. If F is projective as a left and as a right A -module, this is an exact functor preserving finite projectives, hence induces a functor on stable modules compatible with shifts. We always mean these two separate projectivity conditions when describing a bimodule as projective on both sides. They are weaker than projectivity over A^e .

3 From two stable actions to a finite module

The last step of the construction uses a finite bimodule F and a stable map $v : X \rightarrow F \otimes_A X$. Two actions on maps out of X will determine all positive self-extensions of a module over a triangular algebra. We prove this conversion independently of the particular bimodule. The module itself is a finite quotient; complete resolutions are used to compute its extensions.

3.1 The conversion principle

For a finite bimodule F projective on both sides, tensoring with F is exact and preserves projectives. Over a symmetric algebra it therefore induces an exact functor on the stable category and commutes with shifts. In the following statement we write $FX = F \otimes_A X$ and use these shift identifications when writing $F(g)$.

For the triangular algebra $\Lambda = \begin{pmatrix} A & 0 \\ F & A \end{pmatrix}$, we write a left module as a triple (M, N, η) : here M, N are left A -modules and $\eta : F \otimes_A M \rightarrow N$ is A -linear. The action is

$$\begin{pmatrix} a & 0 \\ f & b \end{pmatrix} (m, n) = (am, \eta(f \otimes m) + bn).$$

A morphism (α, β) satisfies $\beta\eta = \eta'(F \otimes \alpha)$.

Proposition 3.1 (Triangular conversion). *Let A be a finite-dimensional symmetric algebra over a field k of characteristic two, let F be a finite A -bimodule projective on both sides, and let X be a finite left A -module. Suppose $v \in \underline{\mathrm{Hom}}_A(X, FX)$. For each integer a , put*

$$H^a = \underline{\mathrm{Hom}}_A(X, X[a]), \quad K^a = \underline{\mathrm{Hom}}_A(X, (FX)[a]),$$

and define

$$\delta^a : H^a \oplus H^a \longrightarrow K^a, \quad (g, j) \longmapsto F(g)v + v[a]j. \quad (3.1)$$

Assume that δ^0 is surjective with nonzero kernel and that δ^a is an isomorphism for every integer $a > 0$.

Choose an actual map $v_0 : X \rightarrow FX$ representing v and an embedding $i : X \rightarrow Q$ into a finite projective A -module. Define

$$\Lambda = \begin{pmatrix} A & 0 \\ F & A \end{pmatrix}, \quad Y = (FX \oplus Q)/\{(v_0(x), i(x)) : x \in X\}, \quad Z = (X, Y, \iota), \quad (3.2)$$

where $\iota : FX \rightarrow Y$ is induced by the first summand and specifies the structure map of the left triangular module. Then Z is a finite nonprojective Gorenstein-projective left Λ -module and

$$\text{Ext}_\Lambda^a(Z, Z) = 0 = \text{Ext}_\Lambda^a(Z, \Lambda) \quad \text{for every integer } a > 0.$$

3.2 Comparison of complete resolutions

For homological complexes $P_\bullet, Q_\bullet$, use the full Hom complex

$$\text{Hom}_R^a(P_\bullet, Q_\bullet) = \prod_{j \in \mathbb{Z}} \text{Hom}_R(P_j, Q_{j-a}), \quad (\partial f)_j = d_{j-a}^Q f_j + f_{j-1} d_j^P. \quad (3.3)$$

This product convention is also used in [21, Section 1.1.1]. The product permits comparison maps with infinitely many nonzero components. Each component of the differential uses only two maps. Its degree- a cocycles are chain maps $P_\bullet \rightarrow Q_\bullet[a]$, and its boundaries are nullhomotopic chain maps. Throughout this section the field has characteristic two, so the displayed signs apply to shifts and cones as well.

Lemma 3.2 (Complete-resolution comparison). *Let R be a finite-dimensional algebra over a field of characteristic two. Let $P_\bullet$ be a totally acyclic complex of finite projective left R -modules, and let $Q_\bullet$ be an exact complex of finite projective left R -modules. Put*

$$C_j(P_\bullet) = \text{coker}(d_{j+1}^P : P_{j+1} \rightarrow P_j), \quad M = C_0(P_\bullet), \quad N = C_0(Q_\bullet).$$

There are natural, composition-compatible identifications

$$H^0 \text{Hom}_R^\bullet(P_\bullet, Q_\bullet) \simeq \underline{\text{Hom}}_R(M, N).$$

Applying this identification to shifted complexes gives, for every integer a ,

$$H^a \text{Hom}_R^\bullet(P_\bullet, Q_\bullet) \simeq \underline{\text{Hom}}_R(C_a(P_\bullet), N) \simeq \underline{\text{Hom}}_R(M, C_{-a}(Q_\bullet)). \quad (3.4)$$

For every integer $a > 0$, these groups are naturally $\text{Ext}_R^a(M, N)$.

Proof. Exactness identifies $C_j(P_\bullet)$ with $\text{im } d_j^P \subset P_{j-1}$. We first record the extension property supplied by total acyclicity: for every finite projective T , a map $\phi : \text{im } d_j^P \rightarrow T$ extends to P_{j-1} . Indeed, $\phi d_j^P : P_j \rightarrow T$ is a cocycle in $\text{Hom}_R(P_\bullet, T)$. Its exactness gives $\psi : P_{j-1} \rightarrow T$ with $\psi d_j^P = \phi d_j^P$, which is the required extension. This argument does not assume that projective R -modules are injective.

A chain map induces a map of cokernels. Conversely, given $u : M \rightarrow N$, first lift $u \in P$ to $f_0 : P_0 \rightarrow Q_0$, where ϵ_P and ϵ_Q denote the quotient maps. Projectivity and exactness give $f_1, f_2, \dots$ successively, with $d_j^Q f_j = f_{j-1} d_j^P$. To extend in the other direction, $d_0^Q f_0$ kills $\text{im } d_1^P = \ker d_0^P$. It therefore induces a map $\text{im } d_0^P \rightarrow Q_{-1}$, which extends to $f_{-1} : P_{-1} \rightarrow Q_{-1}$ by the preceding extension property. The same argument gives $f_{-2}, f_{-3}, \dots$. This constructs a full comparison map.

A nullhomotopy has components $h_j : P_j \rightarrow Q_{j+1}$ and satisfies

$$f_j = d_{j+1}^Q h_j + h_{j-1} d_j^P.$$

On cokernels its first summand vanishes and its second factors through P_{-1} . Thus a nullhomotopic map induces a projective factorization. For the converse, suppose the induced map factors as $M \rightarrow T \rightarrow N$ with T finite projective. Extend $M \rightarrow T$ to P_{-1} by the extension property, and lift $T \rightarrow N$ to Q_0 . Their composite is a homotopy component h_{-1} whose boundary induces exactly the given factorization. Subtract this boundary. It remains to nullhomotope a chain map f' inducing zero on cokernels.

For this remaining map take $h_{-1} = 0$. Since f'_0 has image in $\text{im } d_1^Q$, projectivity gives $h_0 : P_0 \rightarrow Q_1$ with $d_1^Q h_0 = f'_0$. Proceeding upwards, the residual $f'_j + h_{j-1} d_j^P$ lands in $\ker d_j^Q = \text{im } d_{j+1}^Q$ and hence lifts to h_j . Proceeding downwards from $j = -1$, the residual $f'_j + d_{j+1}^Q h_j$ kills $\text{im } d_{j+1}^P$ by the homotopy equation already obtained at $j + 1$. It descends to $\text{im } d_j^P$ and extends to $h_{j-1} : P_{j-1} \rightarrow Q_j$ by the extension property. Each recursion assigns one component at a time and changes only adjacent equations. Together they define a homotopy in the full product (3.3). This proves the degree-zero identification. Passing to cokernels respects composition, and the proved uniqueness modulo homotopy gives its asserted compatibility.

Apply the degree-zero result to $P_\bullet[-a]$ or to $Q_\bullet[a]$ to obtain (3.4). For $a > 0$, $C_a(P_\bullet)$ is the a th syzygy of M . The projective-resolution definition of $\text{Ext}_R^a(M, N)$ quotients $\text{Hom}_R(C_a(P_\bullet), N)$ by maps extending to P_{a-1} . These maps factor through a projective. Conversely, every projective factorization extends to P_{a-1} by the extension property. The two quotients coincide, including when $a = 1$. $\square$

3.3 Proof of the conversion principle

Proof of Proposition 3.1. We first check finiteness and introduce the two column functors. Since i is injective, the map ι in (3.2) is injective and there is an exact sequence

$$0 \longrightarrow FX \xrightarrow{\iota} Y \longrightarrow Q/i(X) \longrightarrow 0. \quad (3.5)$$

In particular all the modules in (3.2) are finite. The algebra is finite as well, with $\dim_k \Lambda = 2 \dim_k A + \dim_k F$.

The column functors below are standard; see [11, Section 2]. Define

$$E(M) = (M, FM, \text{id}), \quad J(N) = (0, N, 0).$$

The functor E is exact because F is projective as a right A -module; J is exact directly. Both preserve finite projectives: $E(A)$ and $J(A)$ are the two projective columns of Λ , and $\Lambda = E(A) \oplus J(A)$ as a left module. The morphism condition above gives

$$\begin{aligned} \text{Hom}_\Lambda(E(M), E(M')) &\simeq \text{Hom}_A(M, M'), \\ \text{Hom}_\Lambda(J(N), J(N')) &\simeq \text{Hom}_A(N, N'), \\ \text{Hom}_\Lambda(E(M), J(N)) &= 0, \\ \text{Hom}_\Lambda(J(N), E(M)) &\simeq \text{Hom}_A(N, FM). \end{aligned} \quad (3.6)$$

In the first line the lower component of a map is $F(\alpha)$. The third line is a vanishing of ordinary module maps, so it holds termwise for complexes in every shift.

Choose a complete resolution $P_\bullet$ of X with $P_{-1} = Q$ and $d_0 = i \epsilon_P$. This is possible by splicing a projective resolution with a projective-injective coresolution, since A is symmetric. Put

$$G_0 = E(P_\bullet), \quad G_1 = J(P_\bullet).$$

They are exact complexes of finite projective Λ -modules. Moreover (3.6) gives

$$\begin{aligned}\mathrm{Hom}_\Lambda(G_0, \Lambda) &\simeq \mathrm{Hom}_A(P_\bullet, A), \\ \mathrm{Hom}_\Lambda(G_1, \Lambda) &\simeq \mathrm{Hom}_A(P_\bullet, F) \oplus \mathrm{Hom}_A(P_\bullet, A).\end{aligned}$$

These complexes are exact: the left A -modules F and A are projective-injective. Thus both G_0 and G_1 are totally acyclic. Also $F \otimes_A P_\bullet$ is a complete resolution of FX . Here right projectivity of F gives exactness, left projectivity gives projective terms, and self-injectivity of A gives Hom exactness.

We can now compute all maps between the two induced complexes. Lemma 3.2 and (3.6) identify their full Hom cohomology as

$$\begin{aligned}H^a \mathrm{End}_\Lambda^\bullet(G_0) &\simeq H^a, & H^a \mathrm{End}_\Lambda^\bullet(G_1) &\simeq H^a, \\ H^a \mathrm{Hom}_\Lambda^\bullet(G_1, G_0) &\simeq K^a, & \mathrm{Hom}_\Lambda^\bullet(G_0, G_1) &= 0.\end{aligned}\tag{3.7}$$

Lift the actual map v_0 to a chain map $V : G_1 \rightarrow G_0$ using the comparison construction, and form

$$P_Z = \mathrm{Cone}(V), \quad (P_Z)_j = (G_0)_j \oplus (G_1)_{j-1}, \quad d_j^{P_Z} = \begin{pmatrix} d_j^{G_0} & V_{j-1} \\ 0 & d_j^{G_1} \end{pmatrix}.$$

The degreewise split cone sequence shows that P_Z is totally acyclic with finite projective terms.

Its cokernel at degree zero is precisely the module Z already defined in (3.2). Its upper component is X . For the lower component, first quotient $F \otimes_A P_0$ by the image of $F \otimes_A d_1$, obtaining FX . The remaining relations, from P_0 , are $(v_0(x), i(x))$: the lift V_0 induces v_0 on cokernels and $d_0 = i\epsilon_P$. The resulting lower component is therefore Y , with structure map ι . This supplies a complete resolution of a concrete finite module.

It remains to compute its self-extensions. Every graded endomorphism of P_Z has two diagonal blocks and one block from $G_1[1]$ to G_0 ; the other direction vanishes by (3.6). Projection to the diagonal gives a short exact sequence of cochain complexes

$$0 \longrightarrow \mathrm{Hom}_\Lambda^\bullet(G_1[1], G_0) \longrightarrow \mathrm{End}_\Lambda^\bullet(P_Z) \longrightarrow \mathrm{End}_\Lambda^\bullet(G_0) \oplus \mathrm{End}_\Lambda^\bullet(G_1[1]) \longrightarrow 0.\tag{3.8}$$

This sequence is split in each graded degree, although the splitting need not commute with the differentials. Reindexing the full product in (3.3) gives

$$H^a \mathrm{Hom}_\Lambda^\bullet(G_1[1], G_0) = K^{a-1}.$$

The connecting map from diagonal degree a to off-diagonal degree $a+1$ is exactly (3.1). Indeed the differential of a diagonal lift has cross term obtained by composing with V on the two sides. The E -diagonal acts there by $F(g)$, and the J -diagonal acts by precomposition with j , as specified in (3.6). Compatibility of comparison with composition identifies this cross term with $F(g)v + v[a]j$.

The long exact sequence of (3.8) consequently contains, for every integer a , the short exact sequence

$$0 \longrightarrow \mathrm{coker} \delta^{a-1} \longrightarrow H^a \mathrm{End}_\Lambda^\bullet(P_Z) \longrightarrow \ker \delta^a \longrightarrow 0.\tag{3.9}$$

For $a=1$, surjectivity of δ^0 and injectivity of δ^1 make both outer terms zero. For $a>1$, the two comparison maps are isomorphisms. Therefore the middle group vanishes for every $a>0$. Lemma 3.2 identifies it with $\mathrm{Ext}_\Lambda^a(Z, Z)$.

At degree zero, (3.9) shows that $H^0 \mathrm{End}_\Lambda^\bullet(P_Z)$ surjects onto the nonzero kernel of δ^0 . By comparison this is a nonzero stable endomorphism group of Z . Its stable identity is

consequently nonzero, since a zero identity would annihilate every stable endomorphism. Thus Z is not projective.

Finally, total acyclicity gives exactness of $\text{Hom}_\Lambda(P_Z, \Lambda)$. Its positive degrees compute the ordinary groups $\text{Ext}_\Lambda^a(Z, \Lambda)$ on the nonnegative resolution tail, so these vanish for every $a > 0$. Exactness also identifies $Z = \text{coker } d_1^{P_Z}$ with the cycle $\text{im } d_0^{P_Z} = \ker d_{-1}^{P_Z}$ in this totally acyclic complex. Hence Z is Gorenstein-projective. $\square$

Proposition 3.1 reduces the construction to a finite bimodule F projective on both sides and a stable map v with the stated two action properties. Surjectivity at degree zero is needed separately: it is the term that removes a possible self-extension in degree one in (3.9).

4 A symmetric algebra and a degree-three class

We begin with a symmetric algebra carrying a simple module and a Hochschild class of degree three. The class will have two uses: its value on the simple will generate the positive self-extension algebra, and an explicit boundary identity will allow us to lift maps through the two cones used later.

4.1 The ten-dimensional algebra and its trivial extension

Let C be the vector space with the following basis, arranged by corners. The notation iCj specifies the left idempotent first.

corner	basis	corner	basis
eCe	e, x, y, z	eCf	u, v
fCe	t, j	fCf	f, n

(4.1)

The elements e, f are orthogonal idempotents with sum 1. Besides their actions specified by the corners, the complete list of nonzero products is

$$\begin{aligned}
 xy = qz, \quad yx = z, \quad xu = yu = v, \quad tx = j, \quad ty = q^2j, \\
 ut = y + qx, \quad vt = qz, \quad uj = z, \quad tu = (1 + q)n, \quad nt = qj, \quad un = v.
 \end{aligned}
 \tag{4.2}$$

All products not specified by these rules are zero.

Lemma 4.1. *The rules (4.1)–(4.2) define an associative unital algebra of dimension ten. Its radical is the span of the eight non-idempotent basis vectors, and*

$$(\text{rad } C)^5 = 0, \quad utut = q(1 + q)z \neq 0.$$

Proof. Give u, t degree one, x, y, n degree two, v, j degree three, and z degree four; the idempotents have degree zero. Every listed product is homogeneous. Associativity involving an idempotent follows from the corners. For three non-idempotent basis vectors, incompatible corners or total degree greater than four force both bracketings to vanish. The complete remaining list is recorded below; direct substitution gives the same value under either bracketing.

abc	utu	tut	xut	yut	ntu	unt	txu	tyu	utx	uty	tun
$(ab)c = a(bc)$	$(1 + q)v$	$q(1 + q)j$	qz	qz	0	qz	0	0	z	q^2z	0

This proves associativity by trilinearity. The positive-degree span N is a nilpotent ideal with $N^5 = 0$ and $C/N \cong k \times k$; therefore $N = \text{rad } C$. Finally, $(ut)u = (1 + q)v$ and $vt = qz$ give the asserted nonzero fourth radical product. $\square$

Remark 4.2. The corner eCe is the quantum exterior algebra $k\langle x, y \rangle / (x^2, y^2, xy - qyx)$, with $z = yx$. The right module fCe has basis t, j and actions $tx = j$, $ty = q^2j$, $jx = jy = 0$. More precisely, put $R = eCe$. The map

$$t \longmapsto x + q^{-1}y, \quad j \longmapsto q^{-1}z$$

identifies fCe with the right ideal $(x + q^{-1}y)R$. In the notation of Schulz [20, Example 7], the parameter is q^{-1} and this ideal is the first syzygy of $(x + y)R$. Here its interactions with the other corners are fixed by the complete multiplication table above.

Put $L = DC$ and equip it with the dual bimodule actions

$$(a\phi)(c) = \phi(ca), \quad (\phi a)(c) = \phi(ac) \quad (a, c \in C, \phi \in L). \quad (4.3)$$

Trivial extensions and their cohomology are studied in [6]. Here the trivial extension is

$$T = C \ltimes L, \quad (a, \phi)(b, \psi) = (ab, a\psi + \phi b). \quad (4.4)$$

For every lower-case basis letter a of C , its upper-case letter a^* denotes the dual basis vector in L ; for example, $j^* = J$ and $f^* = F$. We also set $(a^*)^* = a$. Thus a capital letter J is a basis vector, whereas the radical below is denoted by $\mathfrak{r}$.

Lemma 4.3. *The algebra T is symmetric of dimension twenty. A symmetrizing trace is*

$$\lambda(a, \phi) = \phi(1),$$

and the trace-dual of each basis letter is its starred letter. Its radical $\mathfrak{r}$ is the span of all basis letters except e, f ; it satisfies $\mathfrak{r}^6 = 0$ and $T/\mathfrak{r} \cong k \times k$.

Proof. The dual actions satisfy the bimodule identities by associativity of C , so (4.4) is associative. Moreover,

$$\lambda((a, \phi)(b, \psi)) = \psi(a) + \phi(b),$$

which is symmetric and nondegenerate. In particular, $\lambda(ab^*) = \delta_{a,b}$ for lower-case basis letters a, b . The dual of the corner iCj lies in jLi .

Extend the grading of C by $\deg(a^*) = 5 - \deg(a)$. Formula (4.3) shows that every product in T is homogeneous. All basis vectors except e, f have positive degree, and no degree exceeds five. Their span is therefore a nilpotent ideal whose quotient is $k \times k$; this proves the radical assertions. $\square$

Let s be the one-dimensional left T -module on which f acts as 1 and every other basis letter acts as zero. It is inflated from the corresponding C -simple. Write s^r for the right simple defined by the same character.

4.2 The bar class and the finite identities

Set $I = ke \oplus kf$. The normalized relative bar resolution of T has terms

$$P_n = T \otimes_I \mathfrak{r}^{\otimes n} \otimes_I T \quad (n \geq 0), \quad P_n = 0 \quad (n < 0). \quad (4.5)$$

We write $a_0[a_1] \cdots [a_n]a_{n+1}$ for an elementary tensor. The differential multiplies each adjacent pair of factors and sums the results; the augmentation $P_0 \rightarrow T$ is multiplication. All signs are $+$ because the characteristic is two. Splitting by idempotents expresses each term as a finite sum of projective bimodules $Ti \otimes jT$ with $i, j \in \{e, f\}$. Exactness follows from a contraction: replace the first coefficient a_0 by its projection to $\mathfrak{r}$, insert that projection as a

new first bar, and put 1 in the first coefficient. In augmented degree send a to $1[a]$. Writing σ for this contraction, the identity $d\sigma + \sigma d = \text{id}$ follows by canceling successive multiplications; the I -component of a_0 is balanced over I . This contraction is right T -linear, so $P \otimes_T s$ is a projective resolution of s as well.

Give a lower-case letter weight zero and an upper-case letter weight one. This weight records the effect of scaling the square-zero ideal $L = DC$; it is distinct from the positive grading in Lemma 4.3. The complete table in Appendix A defines an I -bimodule map

$$p : \mathfrak{r} \otimes_I \mathfrak{r} \otimes_I \mathfrak{r} \longrightarrow T.$$

Every unlisted value is zero. The following records exactly the properties of this finite table that the proof needs.

Lemma 4.4. *The cochain p has weight -1 and is a Hochschild cocycle. Its associated class evaluated on s is nonzero in $\text{Ext}_T^3(s, s)$.*

For any $H \in k^\times$, let $h = h_H$ be the automorphism of T fixing C and multiplying L by H . On the radical basis define

$$z_H(a) = \begin{cases} Hqa, & a \in \{u, v, t, E, X, Y, Z, U, V\}, \\ 0, & \text{otherwise,} \end{cases}$$

and extend linearly. For all $a, b \in \mathfrak{r}$ one has

$$\sum_{w \in \mathcal{B}_{\mathfrak{r}}} p(a, b, h(w))w^* = az_H(b) + z_H(ab) + z_H(a)h^{-1}(b), \quad (4.6)$$

where $\mathcal{B}_{\mathfrak{r}}$ is the radical basis.

Proof. The endpoint, weight, cocycle and boundary identities are verified from the complete finite data in Appendix A. We prove the nonzero evaluation explicitly. In the bar complex $s^{\mathfrak{r}} \otimes_T P \otimes_T s$, consider

$$\zeta = q^2[t|x|J] + [t|y|J]. \quad (4.7)$$

Both words have endpoint f on each side. All outer radical actions vanish on the endpoint simples. The multiplication rules, including the dual actions, give

$$tx = j, \quad ty = q^2j, \quad xJ = T, \quad yJ = q^2T.$$

Consequently

$$d\zeta = q^2([j|J] + [t|T]) + q^2[j|J] + q^2[t|T] = 0.$$

The table gives $p(t, x, J) = 0$ and $p(t, y, J) = q^3f$, so its evaluation on ζ is $q^3 \neq 0$. Every T -linear map from $P \otimes_T s$ to s factors through the left simple quotient and hence pairs with this two-sided-simple bar complex. A coboundary pairs to zero with its cycles. Thus the evaluated class is nonzero. $\square$

The evaluation of a Hochschild class on modules is the characteristic action; see [15, Section 4.3]. We use the same letter p for its comparison map $P \rightarrow P[3]$: for $n \geq 3$ apply the cochain to the last three bars and multiply its output into the right coefficient,

$$a_0[a_1 | \cdots | a_n]a_{n+1} \longmapsto a_0[a_1 | \cdots | a_{n-3}]p(a_{n-2}, a_{n-1}, a_n)a_{n+1}; \quad (4.8)$$

the map is zero for $n < 3$. This is a chain map: strict-prefix multiplications cancel, and the remaining equation on the last four bars is precisely the Hochschild cocycle equation. At the bottom index the target differential is zero. Composing its evaluation on s with the augmentation represents the class in Lemma 4.4.

We have now obtained the finite algebra and the specific degree-three class. The next step is to compute all self-extension degrees of s , so that this nonzero class can be identified as a polynomial generator.

5 A simple module with polynomial self-extensions

The cocycle constructed above will control the two cones in our example. We now determine all self-extensions of the simple on which it acts. The key step is a calculation over the smaller algebra C : its simple s has no positive self-extensions, but $\text{Ext}_C^2(s, C)$ is a simple module. Passing to the trivial extension $T = C \ltimes DC$ turns this exceptional class into a polynomial generator of degree three.

5.1 The resolution over C

Let s^r denote the one-dimensional right f -simple. For every integer $i \geq 0$, set

$$\ell_i = x + q^i y.$$

The transcendence of q implies $q \neq 0$ and $1 + q^m \neq 0$ for every positive integer m . We will use these conditions for arbitrarily large indices.

Lemma 5.1. *A projective resolution $R \rightarrow s$ over C has terms*

$$R_0 = Cf, \quad R_i = Ce \quad (i \geq 1),$$

and differentials given by right multiplication:

$$\cdots \longrightarrow Ce \xrightarrow{\cdot \ell_1} Ce \xrightarrow{\cdot \ell_0} Ce \xrightarrow{\cdot u} Cf \longrightarrow s \longrightarrow 0.$$

More precisely, d_{i+2} is right multiplication by ℓ_i for every $i \geq 0$. For every integer $i \geq 0$,

$$\text{Ext}_C^i(s, s) = \begin{cases} k & i = 0, \\ 0 & i > 0, \end{cases} \quad \text{Ext}_C^i(s, C) = \begin{cases} s^r & i = 2, \\ 0 & i \neq 2. \end{cases}$$

The second identification is an identification of right C -modules.

Proof. The left ideals Ce and Cf have respective bases

$$(e, x, y, z, t, j), \quad (f, u, v, n).$$

The multiplication table gives the following formulas, valid for every $i \geq 0$:

a	e	x	y	z	t	j
au	u	v	v	0	$(1+q)n$	0
$a\ell_i$	ℓ_i	$q^{i+1}z$	z	0	$(1+q^{i+2})j$	0

Consequently, right multiplication by u has image $\langle u, v, n \rangle = \text{rad}(Cf)$ and kernel $\langle \ell_0, z, j \rangle$. Right multiplication by ℓ_i has image $\langle \ell_i, z, j \rangle$ and kernel $\langle \ell_{i+1}, z, j \rangle$. In the latter calculation, a kernel vector has zero e - and t -coefficients, while its x, y coefficients satisfy $a_y = q^{i+1}a_x$. These identities prove exactness at every index. Since $\text{Hom}_C(Ce, s) = es = 0$ and $\text{Hom}_C(Cf, s) = fs = k$, they also give the asserted self-extension groups over C .

To compute $\text{Ext}_C^*(s, C)$, identify $\text{Hom}_C(Ce, C) = eC$ and $\text{Hom}_C(Cf, C) = fC$ by evaluation at the idempotent. Precomposition with right multiplication becomes left multiplication. Thus the Hom complex is

$$fC \xrightarrow{\cdot u} eC \xrightarrow{\cdot \ell_0} eC \xrightarrow{\cdot \ell_1} eC \longrightarrow \cdots$$

Left multiplication by u sends the basis (f, t, j, n) of fC to $(u, y + qx, z, v)$. These four vectors are independent. On the basis (e, x, y, z, u, v) of eC , left multiplication by ℓ_i is

$$\begin{array}{c|cccccc} a & e & x & y & z & u & v \\ \hline \ell_i a & \ell_i & q^i z & qz & 0 & (1 + q^i)v & 0. \end{array}$$

The index $i = 0$ is exceptional:

$$\ker(\ell_0 \cdot) = \langle qx + y, z, u, v \rangle = \text{im}(u \cdot), \quad \text{im}(\ell_0 \cdot) = \langle \ell_0, z \rangle.$$

For every $i \geq 1$, the coefficient $1 + q^i$ is nonzero, and

$$\ker(\ell_i \cdot) = \langle \ell_{i-1}, z, v \rangle, \quad \text{im}(\ell_i \cdot) = \langle \ell_i, z, v \rangle.$$

The cohomology in degrees zero and one therefore vanishes. In degree two it is the line represented by v ; in every higher degree the displayed kernel equals the preceding image. Finally, $vf = v$, $ve = 0$, and the only possibly nonzero right radical product is $vt = qz$, which is a boundary. The surviving line is precisely s^τ . $\square$

5.2 The trivial-extension triangle and its product

Write $L = DC$. The duality calculation in Lemma 5.1 determines $L \otimes_C^{\mathbf{L}} s$. The extension $0 \rightarrow L \rightarrow T \rightarrow C \rightarrow 0$ then provides a connecting morphism over T . We identify its products explicitly so that the calculation controls every positive degree.

Lemma 5.2. *For the algebra $T = C \ltimes DC$ and its left f -simple s ,*

$$\text{Ext}_T^*(s, s) = k[\tau], \quad |\tau| = 3,$$

as a graded algebra, where $\tau = p(s)$ is the evaluation of the cocycle from Lemma 4.4.

Proof. For a finite projective left C -module V , evaluation gives a natural isomorphism of left C -modules

$$DC \otimes_C V \longrightarrow D \text{Hom}_C(V, C), \quad \lambda \otimes v \longmapsto (\varphi \longmapsto \lambda(\varphi(v))).$$

Apply this isomorphism termwise to the resolution R of Lemma 5.1. Exactness of vector-space duality shows that $L \otimes_C R$ has homology only in degree two, where it is $D(s^\tau) = s$. Moreover, L acts by zero on the entire complex, since $L^2 = 0$. Good truncations therefore give a T -linear derived identification

$$L \otimes_C R \simeq s[2].$$

Regard $0 \rightarrow L \rightarrow T \rightarrow C \rightarrow 0$ as a sequence of (T, C) -bimodules. Tensoring it termwise with R gives a short exact sequence of complexes of left T -modules,

$$0 \longrightarrow L \otimes_C R \longrightarrow T \otimes_C R \longrightarrow R \longrightarrow 0,$$

where the last complex is inflated along $T \rightarrow C$. Exactness uses projectivity of each left C -module R_i ; it does not require flatness of L or T as right C -modules. Put $\mathcal{I} = T \otimes_C R$. We obtain the derived triangle

$$s[2] \longrightarrow \mathcal{I} \longrightarrow s \xrightarrow{\delta} s[3]. \quad (5.1)$$

The middle complex has homology s in degrees zero and two and no other homology. Its terms are finite dimensional, so good truncation makes it an object of the bounded derived category of finite T -modules.

Each term of $\mathcal{I}$ is a finite projective T -module. Because the complex is bounded below homologically, it computes derived maps out of it: a map from it to an acyclic complex is nullhomotopic by successive projective lifts starting in its bottom degree. Termwise induction and restriction adjunction thus gives, for every integer i ,

$$B^i := \mathrm{Hom}_{D^b(T\text{-mod})}(\mathcal{I}, s[i]) = H^i \mathrm{Hom}_C(R, s) = \begin{cases} k & i = 0, \\ 0 & i \neq 0. \end{cases}$$

Set $E^i = \mathrm{Hom}_{D^b(T\text{-mod})}(s, s[i])$. These are ordinary Ext groups for $i \geq 0$ and vanish for $i < 0$. Applying $\mathrm{Hom}(-, s[i])$ to (5.1) gives the exact segment

$$B^{i-1} \longrightarrow E^{i-3} \xrightarrow{\delta^*} E^i \longrightarrow B^i. \quad (5.2)$$

For $i = 1$, the groups E^{-2} and B^1 vanish, so $E^1 = 0$. For every $i \geq 2$, both displayed B -groups vanish; hence $\delta^* : E^{i-3} \rightarrow E^i$ is an isomorphism. This includes $E^2 = 0$, and at $i = 3$ it sends id_s to the nonzero class δ .

More precisely, if $\beta : s \rightarrow s[i-3]$, its image under this connecting map is

$$\delta^*(\beta) = \beta[3] \circ \delta.$$

Thus (5.2) is multiplication by δ in the Yoneda algebra. Every power δ^m is nonzero and spans E^{3m} , and all other nonnegative degrees vanish. We have proved $\mathrm{Ext}_T^*(s, s) = k[\delta]$ as an algebra.

Lemma 4.4 shows that $p(s)$ is nonzero in degree three: the simple-valued cochain evaluates to $q^3 \neq 0$ on the cycle $q^2[t|x|J] + [t|y|J]$. Hence $p(s)$ is a nonzero scalar multiple of δ and is itself a polynomial generator. This proves the stated normalization $\tau = p(s)$ without choosing an isomorphism $L \otimes_C R \simeq s[2]$ with a prescribed scalar. $\square$

5.3 Two independent degree-three generators

We now pass to the symmetric algebra and simple used in the remaining construction:

$$A = T \otimes_k T, \quad X = s \otimes_k s.$$

The tensor product of the symmetrizing forms makes A symmetric, and $\dim_k A = 400$. Let p_1, p_2 be the external copies of the bimodule class p on the two factors, and write $\tau_i = p_i(X)$ for their evaluations.

Lemma 5.3. *The simple left A -module X is nonprojective, and*

$$\bigoplus_{a \geq 0} \underline{\mathrm{Hom}}_A(X, X[a]) = k[\tau_1, \tau_2], \quad |\tau_1| = |\tau_2| = 3.$$

The multiplication is composition with shifts, and its positive-degree part agrees with the ordinary Ext algebra.

Proof. Tensor two projective T -resolutions of s over k . In each total degree the direct sum is finite, so this is a resolution of X by finite projective A -modules. The Hom complex is the total tensor product of the two Hom complexes over T . Splitting complexes of vector spaces gives the Künneth isomorphism. Tensoring the comparison maps identifies multiplication as well; the external maps commute in characteristic two. Lemma 5.2 therefore gives

$$\mathrm{Ext}_A^*(X, X) = k[\tau_1, \tau_2].$$

In particular X has a nonzero positive self-extension and cannot be projective. Its ordinary endomorphism ring is k , since X is one dimensional. Its identity cannot factor through a projective module: such a factorization would make X a direct summand of that projective. Thus its stable endomorphism ring is also k . Positive stable Hom agrees with Ext over the symmetric algebra A , giving the result. $\square$

We have obtained two polynomial classes acting on X , together with specific bimodule representatives for them. The next step uses their cones to leave only two stable Hom groups; their degrees will determine the shift in the subsequent fiber construction.

6 Two cones with two surviving Hom groups

We now use the two degree-three classes to construct an A -bimodule $\mathcal{C}$, projective on each side, for which the stable maps from X to $\mathcal{C} \otimes_A X[a]$ survive only in degrees 0 and 3. The degree-three group will detect the maps constructed in the next section. Throughout this section, put

$$A = T \otimes_k T, \quad X = s \otimes_k s, \quad R = A^e.$$

The algebras A and R are symmetric: their symmetrizing forms are the tensor products of the form on T and, for opposite factors, the same form. Write

$$H^a = \underline{\text{Hom}}_A(X, X[a]).$$

The preceding calculation gives

$$H^{\geq 0} = k[\tau_1, \tau_2], \quad |\tau_1| = |\tau_2| = 3. \quad (6.1)$$

In degree zero this is also the stable endomorphism space: X is one-dimensional, and its nonzero degree-three self-extension proves that it is nonprojective.

6.1 Negative degrees and multiplication

We first record the symmetric stable duality, including the compatibility with multiplication that will be needed below; see also [14, Section 2]. For finite modules M, N over a symmetric algebra G , it says that

$$D\underline{\text{Hom}}_G(M, N) \simeq \underline{\text{Hom}}_G(N, M[-1]), \quad D = \text{Hom}_k(-, k), \quad (6.2)$$

naturally in N . Here is a direct proof with our indexing conventions. Choose a complete resolution Q of M , so that $M = \text{coker}(Q_1 \rightarrow Q_0)$. For a finite projective Q_i , symmetry gives a natural isomorphism

$$D \text{Hom}_G(Q_i, N) \simeq \text{Hom}_G(N, Q_i).$$

It suffices to check this for $Q_i = G$, where it is the isomorphism $DG \simeq G$, and then to pass to finite sums and summands. Dualizing the cochain complex $\text{Hom}_G(Q, N)$ therefore gives the chain complex $\text{Hom}_G(N, Q)$. Its homology at zero consists of maps into $\ker(Q_0 \rightarrow Q_{-1}) = \Omega M$, modulo maps through Q_1 . These are exactly the projective factorizations, since maps from a projective to ΩM lift through $Q_1 \twoheadrightarrow \Omega M$. Vector-space duality is exact, proving (6.2) and its naturality.

In particular,

$$DH^a \simeq H^{-a-1}. \quad (6.3)$$

For $m \geq 0$, let $\mathcal{P}_m$ denote the degree- m part of the polynomial ring $k[\tau_1, \tau_2]$ when each variable is assigned degree one. Thus

$$H^{3m} = \mathcal{P}_m, \quad H^{-3m-1} = D\mathcal{P}_m \quad (m \geq 0), \quad (6.4)$$

and all other groups vanish.

These identifications also determine the action of the two classes in negative degrees. Postcomposition by $\tau_i[a]$ gives $H^a \rightarrow H^{a+3}$. By naturality of (6.2), its transpose, after shifting the source back to X , is

$$H^{-a-4} \longrightarrow H^{-a-1}, \quad g \longmapsto g[3] \circ \tau_i.$$

When $a \leq -4$, the last map is multiplication by τ_i in the polynomial ring (6.1). The action within negative degrees is therefore the dual of polynomial multiplication. There is no nonzero product crossing from negative to nonnegative degree: the only nonzero possible source is H^{-1} , and its target is $H^2 = 0$.

6.2 Finite bimodules from projective tails

We will form the two cones as ordinary complexes before passing to stable bimodules. We give the passage explicitly to retain finite representatives and exact tensor evaluation.

Bimodules attached to Hochschild classes and their iterated tensor products have a precursor in the support-variety constructions of Erdmann, Holloway, Snashall, Solberg and Taillefer [10, Section 2]. Here we compute the full stable Hom profile directly, including negative degrees and the top attaching map.

If V is a bounded below complex of finite projective R -modules that is exact in sufficiently high degrees, put

$$\langle V \rangle = (\text{coker}(V_{N+1} \longrightarrow V_N))[N] \quad (6.5)$$

in the stable category, with N sufficiently large. This does not depend on N : writing V_N^c for this cokernel, exactness gives

$$0 \longrightarrow V_{N+1}^c \longrightarrow V_N \longrightarrow V_N^c \longrightarrow 0.$$

Thus $V_{N+1}^c \simeq V_N^c[-1]$ stably. Chain maps defined in sufficiently high degrees induce maps on these cokernels, compatibly with increasing N and with shifts. A chain homotopy changes the bottom induced map by a factorization through a projective, so homotopic tail maps induce the same stable map.

For a degreewise split short exact sequence of such complexes $0 \rightarrow B \rightarrow V \rightarrow D \rightarrow 0$, their cokernels at N form a short exact sequence once $H_{N+1}(D) = 0$. Indeed, a possible extra relation in the B -cokernel is obtained by lifting a cycle in D_{N+1} . That cycle is a boundary; subtracting a lifted boundary leaves a relation already in B . The resulting stable sequence has the usual long exact stable Hom sequence: apply Hom from a complete projective resolution to the short exact sequence and use its cohomology description of stable Hom. If the cross term of the differential is $r : D_j \rightarrow B_{j-1}$, the connecting stable map $\langle D \rangle \rightarrow \langle B \rangle[1]$ is induced by r . This follows by lifting along the degreewise splitting and taking the B -component of the differential, which is precisely r .

Call a bimodule *side-projective* if it is projective separately as a left and as a right A -module. Finite stable shifts of a finite side-projective bimodule have finite side-projective representatives. For a syzygy, take a finite R -projective surjection; its kernel sequence splits on each side because its quotient is side-projective. For a cosyzygy, take a finite R -projective-injective embedding; its cokernel sequence splits on each side because its submodule is side-injective, by symmetry of A . Likewise a stable map $M \rightarrow N$ between such bimodules

has a finite side-projective cone: represent the map by f , embed M into a finite R -projective I , and take the cokernel of $(f, i) : M \rightarrow N \oplus I$. Again the sequence splits on each side.

Lemma 6.1. *Let P be the bimodule bar resolution of T and let $p : P \rightarrow P[3]$ be the selected comparison map. Form the ordinary complex*

$$K = \text{Cone}(p : P[-3] \rightarrow P) \otimes_k \text{Cone}(p : P[-3] \rightarrow P).$$

Then K has finite projective R -terms and is exact in degrees greater than zero. For every $N \geq 1$, its cokernel $C_N = \text{coker}(K_{N+1} \rightarrow K_N)$ is finite and side-projective. Consequently $C = \langle K \rangle$ has a finite side-projective module representative. The same conclusions hold for the first two filtration levels of K .

Proof. Write $P^{\text{tot}} = P \otimes_k P$. In each total degree there are only finitely many summands, each an external tensor of projective bimodules; these are projective over R . The complex P^{tot} resolves A . Label the four cone cells by subsets $S \subseteq \{1, 2\}$, where $i \in S$ means that the shifted cell is selected in cone factor i . The S -cell is $P^{\text{tot}}[-2|S|]$; its differential to the cell $S \setminus \{i\}$ is the shifted comparison map p_i . Filtering by $|S|$ gives the three associated levels

$$P^{\text{tot}}, \quad (P^{\text{tot}}[-2])^{\oplus 2}, \quad P^{\text{tot}}[-4]. \quad (6.6)$$

The cell homology is supported at $0, -2, -4$, respectively. The filtration long exact sequences therefore show that K is exact in positive degrees.

For $N \geq 1$, the cokernel filtration is exact by the preceding observation. Its factors, with multiplicities $1, 2, 1$, are

$$\text{coker}(P_{N+2r+1}^{\text{tot}} \rightarrow P_{N+2r}^{\text{tot}}), \quad r = 0, 1, 2.$$

The augmented resolution of A splits on either A -side, since the regular module is projective; inductively all its syzygies are projective on each side. Thus these filtration factors are side-projective. Their extensions split on each side, proving the same property for C_N . Finally $C_N[N]$ has a finite side-projective representative by the shift construction above. In particular one may take $N = 1$ and one finite cosyzygy of C_1 . $\square$

Forming the external tensor of ordinary complexes gives all four cells of (6.6) as projective resolutions of shifts of the regular A -bimodule. We pass to their stable representatives only after this construction.

Evaluation on a finite left module means tensoring over A on the right. Tensoring with a side-projective bimodule is an exact functor of left A -modules and preserves projectives. Every R -projective evaluates to an A -projective, since a free bimodule evaluates to a module of the form $A \otimes_k V$. For evaluation at a fixed module, the right-split sequences used above remain exact. Their evaluations therefore preserve the stable shifts and long exact Hom sequences in the construction. Accordingly the filtration (6.6), after evaluation on X , has stable cells

$$X, \quad X[-2]^{\oplus 2}, \quad X[-4].$$

Its adjacent connecting maps are the shifted actions of τ_1, τ_2 : they are induced by the original bimodule comparison maps tensored with X . This specifies the attaching maps whose Hom groups we now compute.

6.3 The Hom profile and its top projection

Proposition 6.2. *For the finite side-projective bimodule $\mathcal{C}$ of Lemma 6.1, put*

$$W^a = \underline{\text{Hom}}_A(X, (\mathcal{C} \otimes_A X)[a]).$$

For every integer a ,

$$W^a = \begin{cases} k, & a = 0 \text{ or } 3, \\ 0, & a \notin \{0, 3\}. \end{cases}$$

Moreover the projection to the top cell induces an isomorphism

$$W^3 \xrightarrow{\sim} H^{-1}. \quad (6.7)$$

Proof. Let $\mathcal{C}_{\leq 1}$ denote the first two filtration levels, represented by their projective tails as above. In total cohomological degree a , the Hom groups of the three associated levels are

$$B^a = H^a, \quad D^a = (H^{a-2})^2, \quad E^a = H^{a-4}.$$

The adjacent connecting maps are

$$\begin{aligned} d_a : D^a &\longrightarrow B^{a+1}, & d_a(u, v) &= \tau_1 u + \tau_2 v, \\ d'_a : E^a &\longrightarrow D^{a+1}, & d'_a(z) &= (\tau_2 z, \tau_1 z). \end{aligned}$$

The order in the column records that deleting the first slot from the full cell leaves the second singleton, and conversely.

We need the degree-wise exact Koszul sequence

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{(\tau_2, \tau_1)^t} \mathcal{P}_m^{\oplus 2} \xrightarrow{(\tau_1, \tau_2)} \mathcal{P}_{m+1} \longrightarrow 0 \quad (m \geq 0), \quad (6.8)$$

where $\mathcal{P}_j = 0$ for $j < 0$. The row is surjective because every positive-degree monomial is divisible by one of the variables. The column is injective, and coprimality of the variables shows that every relation $\tau_1 u + \tau_2 v = 0$ is of the form $(u, v) = (\tau_2 w, \tau_1 w)$. On negative groups, dualize (6.8) and interchange its two middle components. By the multiplication calculation above this gives exactly the sequence through D^a there. At the seam its terminal zero space is the actual group $H^2 = 0$. We obtain, in all degrees,

$$\begin{aligned} \ker d_a &= \text{im } d'_{a-1}, \\ \text{coker } d_{a-1} &= \begin{cases} k, & a = 0, \\ 0, & a \neq 0, \end{cases} & \ker d'_a &= \begin{cases} k = H^{-1}, & a = 3, \\ 0, & a \neq 3. \end{cases} \end{aligned} \quad (6.9)$$

For the column kernel, its maps on $D\mathcal{P}_m$ for $m \geq 1$ are dual to the surjective positive row and hence injective. Its remaining source is $D\mathcal{P}_0 = H^{-1}$, whose two products vanish in H^2 .

It remains to check that the actual filtration has no further effect on these scalar classes. Put

$$K_1^a = \underline{\text{Hom}}_A(X, (\mathcal{C}_{\leq 1} \otimes_A X)[a]).$$

The first filtration long exact sequence gives

$$0 \longrightarrow \text{coker } d_{a-1} \longrightarrow K_1^a \xrightarrow{q_a} \ker d_a \longrightarrow 0. \quad (6.10)$$

The top-cell connecting map is $r_a : E^a \rightarrow K_1^{a+1}$, and its projection satisfies $q_{a+1} r_a = d'_a$. Thus $\ker r_a \subseteq \ker d'_a$, so it can be nonzero only when $a = 3$. At that index $K_1^4 = 0$ by (6.10), since $D^4 = (H^2)^2 = 0$. Hence $\ker r_3 = E^3 = H^{-1}$.

The image of r_{a-1} surjects onto $\ker d_a$ by (6.9). When $a \neq 0$, the left term of (6.10) vanishes, so r_{a-1} is surjective. When $a = 0$, that sequence gives $K_1^0 = k$, whereas the only incoming source is $E^{-1} = H^{-5} = 0$. Therefore $\operatorname{coker} r_{a-1}$ is k exactly at $a = 0$ and zero elsewhere.

Finally the second filtration long exact sequence gives

$$0 \longrightarrow \operatorname{coker} r_{a-1} \longrightarrow W^a \longrightarrow \ker r_a \longrightarrow 0.$$

This proves the asserted groups. At $a = 3$ its left term vanishes and its right term is all of $E^3 = H^{-1}$, so the actual top-cell projection is an isomorphism, proving (6.7). $\square$

The two exact sequences in the proof account for the complete filtration, including any attaching map not determined by its adjacent component. We have therefore obtained the required two-scalar profile together with a concrete way to detect its degree-three class: after shifting the cone by three, it suffices to project to $X[-1]$. The next construction supplies maps with a nonzero projection there.

7 Twisted bimodules and a lift to the two cones

We now construct two stable bimodule maps whose evaluations generate the one-dimensional degree-three group of the cone calculation. Their sources will have distinct twisting actions. The essential point is a chain-level lift: we give the homotopies in every degree and identify its projection to the top cell.

For a nonzero scalar $H \in k$, let $h = h_H$ be the automorphism of T that fixes C and multiplies DC by H . Write T_h for the bimodule with its usual left action and right action $x \cdot a = xh(a)$. Put

$$U = T_h, \quad Q = P_h, \quad U_H = A_{h \otimes h}, \quad Q^{\text{tot}} = Q \otimes_k Q.$$

Thus Q is the right-twisted bar resolution of U , with augmentation $\epsilon : Q_0 \rightarrow U$. A formula on its bar generators extends by

$$\psi(a_0[a_1] \cdots [a_i]a_{i+1}) = a_0\psi([a_1] \cdots [a_i])h^{-1}(a_{i+1}). \quad (7.1)$$

Here the displayed terminal coefficient is its underlying element of T . This convention makes ψ right linear from the twisted source to an untwisted target.

Proposition 7.1. *For each $H \in k^\times$, there is a stable bimodule map*

$$f_H : U_H \longrightarrow \mathcal{C}[3]$$

whose composite with the top-cell projection $\mathcal{C}[3] \rightarrow A[-1]$ is represented by the bimodule map $\beta_H : U_H \rightarrow \Omega A \subset P_0 \otimes P_0$ determined by

$$\beta_H(1) = \left(\sum_w h(w) \otimes_I w^* \right) \otimes \left(\sum_w h(w) \otimes_I w^* \right),$$

where each sum is over the letter basis of T . Under the canonical identification $U_H \otimes_A X \simeq X$, its projection after evaluation sends

$$1 \longmapsto H^2(f^* \otimes f^*) \quad \text{in} \quad \Omega X = \operatorname{rad}(A)(f \otimes f) \subset A(f \otimes f).$$

This evaluated map is nonzero modulo maps factoring through projectives. In particular, $f_H X$ is nonzero in $\underline{\operatorname{Hom}}_A(X, \mathcal{C}X[3])$.

7.1 A twisted trace-dual tensor

For a basis letter w of T , write w^* for its trace-dual letter. In $P_0 = T \otimes_I T$, set

$$\xi_H = \sum_{w \text{ in the basis of } T} h(w) \otimes_I w^*. \quad (7.2)$$

The trace-dual coevaluation identity is standard; see [4, Proposition 3.3]. The associative symmetric trace gives the ordinary coevaluation identity $a \sum_w w \otimes w^* = \sum_w w \otimes w^* a$. Applying h to the first factor and projecting to $\otimes_I$ gives

$$h(a)\xi_H = \xi_H a.$$

Consequently $b : U \rightarrow P_0$, defined by $b(x) = x\xi_H$, is a bimodule map.

We need the stronger fact that b takes values in $\ker \pi$, where $\pi : P_0 \rightarrow T$ is multiplication. For a lower-case basis letter w of C , the product ww^* is the dual of its left vertex idempotent, and w^*w is the dual of its right vertex idempotent. Indeed $(ww^*)(c) = w^*(cw)$; the positive grading makes this zero for $c \in \text{rad } C$, and the idempotent values determine the assertion. The row dimensions of C are 6, 4, and its column dimensions are also 6, 4. Separating the lower-case and upper-case letters therefore gives

$$\pi\xi_H = \sum_{w \text{ lower-case}} ww^* + H \sum_{w \text{ lower-case}} w^*w = 0$$

in characteristic two. The same calculation on each left vertex $r \in \{e, f\}$ gives

$$\sum_{\substack{w \in \mathfrak{r} \text{ a basis letter} \\ \text{left}(w)=r}} h(w)w^* = r^*. \quad (7.3)$$

Here the full sum is zero and the omitted idempotent term is $rr^* = r^*$.

Taking the external square, define

$$\beta_H = b \otimes_k b : U_H \longrightarrow \ker(P_0^{\text{tot}} \rightarrow A) = \Omega A, \quad P^{\text{tot}} = P \otimes_k P. \quad (7.4)$$

Its image lies in the indicated kernel because $\pi b = 0$. The remaining task is to lift this map through the top-cell projection.

7.2 Two homotopies on the bar resolution

Let $D_b : Q \rightarrow P$ be $b\epsilon$ in degree zero and zero in every positive degree. It is a chain map. A homotopy for D_b will supply a comparison representing β_H in the top cell. Define maps $B_i : Q_i \rightarrow P_{i+1}$ for every $i \geq 0$ by

$$B_i([a_1 | \cdots | a_i]) = \sum_{w \in \mathfrak{r} \text{ a basis letter}} [a_1 | \cdots | a_i | h(w)]w^*, \quad (7.5)$$

retaining only composable terms and extending by (7.1).

Lemma 7.2. *On the nonnegative resolutions, with $B_{-1} = 0$, one has*

$$dB + Bd = D_b.$$

Proof. For $a \in \mathfrak{r}$ with right vertex r , project the first factor of $a\xi_H = \xi_H h^{-1}(a)$ to $\mathfrak{r}$ along I . Removing the idempotent terms leaves

$$\sum_{w \in \mathfrak{r}} ah(w) \otimes w^* + \sum_{w \in \mathfrak{r}} h(w) \otimes w^* h^{-1}(a) = a \otimes r^*, \quad (7.6)$$

with the appropriate vertex restrictions in each sum. On the left side of the original identity, the omitted term is $a \otimes r^*$; on its right side the omitted first factors lie in I and disappear under the projection.

For $i > 0$, expand $dB_i + B_{i-1}d$ on a bar generator. All multiplications strictly before the old final bar cancel in pairs. The two terms adjacent to the old final letter have sum $[a_1 | \cdots | a_i]r^*$ by (7.6). The last multiplication in dB_i is exactly the same expression by (7.3), so these terms also cancel.

At degree zero, on the generator $r[]r$, the differential gives

$$dB_0(r[]r) = \sum_{\substack{w \in \mathfrak{r} \\ \text{left}(w)=r}} h(w)[]w^* + r[] \sum_{\substack{w \in \mathfrak{r} \\ \text{left}(w)=r}} h(w)w^*.$$

The second term is $r[]r^*$ by (7.3). It restores the omitted idempotent term of the full tensor (7.2); the result is $r\xi_H = b\epsilon(r[]r)$. Bimodule linearity proves the identity on every degree-zero element. $\square$

We now use B to represent the prescribed top projection. In $K[3]$, the cell indexed by $\{1, 2\}$ is $P^{\text{tot}}[-1]$. Put $V = B \otimes D_b$, which raises the underlying total degree by one, as this shift requires. Lemma 7.2 gives

$$dV + Vd = (b \otimes b)\epsilon^{\text{tot}},$$

where the right side is supported in degree zero. Thus $dV_0 = \beta_H \epsilon_0^{\text{tot}}$, and at positive indices V satisfies the comparison equation. The complex with terms P_{i+1}^{tot} and augmentation to ΩA resolves ΩA . Consequently V is a comparison lifting β_H .

To extend V from the top cell to $K[3]$, we must account for the two attaching maps. Deleting slot 1 contributes $pB \otimes D_b$ to the cell indexed by $\{2\}$. The suffix comparison $p : P \rightarrow P[3]$ from the finite cochain data lowers the underlying bar degree by three, so $pD_b = 0$. The contribution from deleting slot 2 is therefore zero. It remains to cancel $pB \otimes D_b$ by a component in cell $\{2\}$. Lemma 7.2 shows that $pB : Q \rightarrow P[2]$ is a chain map. The additional boundary identity (4.6) supplies its nullhomotopy; cocycle closure by itself would not suffice.

Lemma 7.3. *There are bimodule maps $G_i : Q_i \rightarrow P_{i-1}$ for all $i \geq 0$ such that*

$$pB = dG + Gd.$$

Proof. Let $z' : Q_1 \rightarrow T$ be the cochain given by $z'([a]) = z_H(a)$ from Lemma 4.4, with extension (7.1). Its boundary on $[a|b]$ is

$$az'(b) + z'(ab) + z'(a)h^{-1}(b).$$

By identity (4.6), this equals $\sum_{w \in \mathfrak{r}} p(a, b, h(w))w^*$, which is the augmentation of $(pB)_2$. Hence

$$\pi(pB)_2 = z'd : Q_2 \longrightarrow T. \quad (7.7)$$

Set $G_0 = 0$ and choose $G_1 : Q_1 \rightarrow P_0$ with $\pi G_1 = z'$. The choice is possible because Q_1 is projective and π is surjective. By (7.7), the map $(pB)_2 + G_1d$ has image in $\ker \pi = \text{im}(d : P_1 \rightarrow P_0)$. Projectivity of Q_2 therefore supplies G_2 with $dG_2 = (pB)_2 + G_1d$.

Suppose $i \geq 3$ and the homotopy equation has been established below i . Since pB is a chain map,

$$d((pB)_i + G_{i-1}d) = (pB)_{i-1}d + ((pB)_{i-1} + G_{i-2}d)d = 0.$$

Exactness of P in positive degree and projectivity of Q_i allow this map to be lifted through $d : P_{i-1} \rightarrow P_{i-2}$, giving G_i . This proves the equation for every $i \geq 2$. At $i = 0, 1$ both sides are zero, since pB has negative target index and $G_0 = 0$. $\square$

7.3 The tail map and its evaluation

The homotopy G supplies the required correction in the singleton cell $\{2\}$, whose shift in $K[3]$ is 1. Together with the top component V , it defines a graded map $\Phi : Q^{\text{tot}} \rightarrow K[3]$:

$$\Phi_{\{1,2\}} = B \otimes D_b, \quad \Phi_{\{2\}} = G \otimes D_b, \quad \Phi_{\{1\}} = \Phi_{\emptyset} = 0. \quad (7.8)$$

The second component lowers underlying degree by one, as the singleton shift requires; the first is the comparison V constructed above.

In the top cell, the chain defect $d\Phi + \Phi d$ is $D_b \otimes D_b$. In cell $\{2\}$, deletion of slot 1 gives $pB \otimes D_b$, canceled by $(dG + Gd) \otimes D_b$. Deleting slot 2 from either nonzero component gives zero because $pD_b = 0$. Thus the only defect is $D_b \otimes D_b$, supported in total source degree zero. The map therefore commutes with the differential in every positive total degree. Passing to high cokernels, at a common index at least 4 as in Lemma 6.1, gives $f_H : U_H \rightarrow \mathcal{C}[3]$.

The top component is still V , so the resulting stable projection is β_H , as identified above.

Lemma 7.4. *Under $U_H \otimes_A X \simeq X$, the evaluated map $\beta_H X$ sends 1 to $H^2(f^* \otimes f^*)$ in $\Omega X \subset A(f \otimes f)$. It does not factor through a projective A -module.*

Proof. In one slot, tensoring (7.2) with the simple leaves only the term whose right coefficient is f . It is the term $w = f^*$, so its value is Hf^* . Taking the external square gives the stated value. The cover sequence remains exact on evaluation because it splits on the right. Its evaluated degree-zero term is $A(f \otimes f)$, the projective cover of X , with kernel $\text{rad}(A)(f \otimes f)$.

Suppose a map $X \rightarrow \text{rad}(A)(f \otimes f)$ factors through a projective. It then factors through a finite free module. Every component of a map $X \rightarrow A$ has image in the left socle, since $\text{rad}(A)X = 0$. In a symmetric algebra the left socle also annihilates the radical on the right: if $\text{rad}(A)u = 0$, then, for $j \in \text{rad}(A)$ and $a \in A$,

$$\text{tr}(auj) = \text{tr}(jau) = 0,$$

so nondegeneracy of the trace gives $uj = 0$. A map $A \rightarrow \text{rad}(A)(f \otimes f)$ is right multiplication by its value at 1, which lies in $\text{rad}(A)$, and hence kills the left socle. Every such factorization is therefore zero. Our displayed map has the nonzero value $H^2(f^* \otimes f^*)$, proving the claim. $\square$

By Proposition 6.2, the top-cell projection is an isomorphism on the degree-three group of the cone calculation. Lemma 7.4 therefore proves that $f_H X$ is nonzero, completing Proposition 7.1. We have obtained the two required maps by taking $H = H_1, H_2$; the next section combines them into a finite bimodule.

8 Two twists and a finite bimodule fiber

We now construct the finite bimodule and the stable map required by Proposition 3.1. The two twists will act differently on every positive self-extension of X , while agreeing on its identity. The fiber of their maps to $\mathcal{C}[3]$ turns this distinction into the comparison isomorphisms used in that proposition.

8.1 A finite representative and an exact sequence

Write $R = A^e$. Lemma 6.1 provides the finite bimodule

$$C_1 = \text{coker}(K_2 \longrightarrow K_1), \quad \mathcal{C} = C_1[1]$$

in the stable category of R . It is projective on both A -sides. To fix actual finite representatives for its shifts, for a finite left R -module M use the injection

$$j_M : M \longrightarrow I_R(M) := \text{Hom}_k(R, M), \quad j_M(m)(t) = tm, \quad (r\varphi)(t) = \varphi(tr). \quad (8.1)$$

Evaluation at 1 proves injectivity. Since R is symmetric, $I_R(M) \cong DR \otimes_k M \cong R \otimes_k M$ as left R -modules, so this is a finite projective module. Put $\Sigma_R M = \text{coker } j_M$. When M is projective on each A -side, the sequence defining $\Sigma_R M$ splits on each side: A is self-injective, so M is injective on either side. Thus $\Sigma_R M$ remains projective on both sides and represents $M[1]$.

In particular, fix the actual finite bimodule

$$\mathcal{Y} = \Sigma_R^4 C_1. \quad (8.2)$$

It is projective on both sides and represents $\mathcal{C}[3]$. This fixes one target for both lifts of Proposition 7.1. Concretely, the tail map there at index 4 induces $\Omega_R^4 U_H \rightarrow C_1$, because

$$\text{coker}(K[3]_5 \longrightarrow K[3]_4) = \text{coker}(K_2 \longrightarrow K_1).$$

Taking four cosyzygies gives the stable map $U_H \rightarrow \mathcal{Y}$, using $\Sigma_R^4 \Omega_R^4 U_H \cong U_H$ stably. It therefore has an actual bimodule-map representative between these finite modules.

For $i = 1, 2$, put

$$\sigma_i = h_{H_i} \otimes h_{H_i}, \quad U_i = A_{\sigma_i}.$$

Thus U_i has its usual left action and right action $u \cdot a = u\sigma_i(a)$. Each U_i is free of rank one on both A -sides. The canonical identification $U_i \otimes_A X = X$ uses the fact that σ_i fixes the simple character of X .

By Lemma 7.4, the top projections of the evaluated lifts are $H_i^2 \beta_0$ for one fixed nonzero class $\beta_0 \in \underline{\text{Hom}}_A(X, X[-1])$. By Proposition 6.2, the top projection

$$W^3 = \underline{\text{Hom}}_A(X, \mathcal{C}X[3]) \longrightarrow \underline{\text{Hom}}_A(X, X[-1])$$

is an isomorphism. Rescale the two lifts by H_i^{-2} , and choose actual representatives $g_i : U_i \rightarrow \mathcal{Y}$ of these rescaled maps. Their evaluations are the same nonzero element $w \in W^3$.

Take the finite free R -module $Q = R \otimes_k \mathcal{Y}$ and its surjection $\pi : Q \rightarrow \mathcal{Y}$, $\pi(r \otimes y) = ry$. Define F by the exact sequence

$$0 \longrightarrow F \longrightarrow U_1 \oplus U_2 \oplus Q \xrightarrow{(g_1, g_2, \pi)} \mathcal{Y} \longrightarrow 0, \quad (8.3)$$

and let $\rho_i : F \rightarrow U_i$ be the two projections. This is a construction of finite modules: C_1 is a cokernel of two finite terms, four finite cokernels give $\mathcal{Y}$, and one finite kernel gives F . Since $\mathcal{Y}$ is projective on each side, (8.3) splits on each side. Consequently F is projective as both a left and a right A -module.

Tensoring (8.3) with X preserves exactness. Furthermore, $Q \otimes_A X$ is projective over A : evaluation of a free A^e -module is a free A -module, and evaluation preserves direct summands. Thus the evaluated stable triangle is

$$FX \longrightarrow X \oplus X \xrightarrow{(w, w)} \mathcal{C}X[3] \longrightarrow FX[1]. \quad (8.4)$$

Here and below FM means $F \otimes_A M$. Tensoring with F is exact because F is right projective; it preserves projectives because F is left projective. It therefore acts on stable maps compatibly with shifts. The actual maps ρ_i give natural transformations from this functor to tensoring with U_i .

Lemma 8.1. *Put $V^a = \underline{\text{Hom}}_A(X, FX[a])$ for each integer a . There is a unique $v \in V^0$ whose two projections are id_X . It spans V^0 . For every integer $a > 0$, projection induces an isomorphism*

$$(\rho_1[a], \rho_2[a]) : V^a \xrightarrow{\sim} (H^a)^2, \quad H^a = \underline{\text{Hom}}_A(X, X[a]). \quad (8.5)$$

Proof. Applying stable Hom from X to (8.4) gives

$$(H^{a-1})^2 \longrightarrow W^{a+2} \longrightarrow V^a \longrightarrow (H^a)^2 \longrightarrow W^{a+3}.$$

At $a = 0$, the first relevant group is $W^2 = 0$, and the last map is $k^2 \rightarrow W^3$, $(c, d) \mapsto (c + d)w$. It is surjective, so V^0 identifies with the diagonal line in k^2 . This proves the assertion about v .

At $a = 1$, the preceding map $(H^0)^2 \rightarrow W^3$ is precisely this surjection. It kills the potential connecting term from W^3 to V^1 , and $W^4 = 0$. Therefore (8.5) is an isomorphism in degree 1 as well. For $a \geq 2$, both W^{a+2} and W^{a+3} vanish by Proposition 6.2, giving the isomorphism directly. $\square$

We have obtained an actual finite bimodule F and a nonzero stable map $v : X \rightarrow FX$. It remains to compare tensoring an extension with F and composing that extension with v .

8.2 The twist eigenvalues and the comparison map

Lemma 8.2. *For $H \in \{H_1, H_2\}$, tensoring with $U_H = A_{h_H \otimes h_H}$, under its canonical identification on X , acts on the whole space H^{3m} as multiplication by H^{-m} for every integer $m \geq 0$.*

Proof. The inverse automorphism in this formula follows from the right-twist convention. For an automorphism h of T and a left T -module M , the balanced map

$$T_h \otimes_T M \longrightarrow {}_{h^{-1}}M, \quad a \otimes m \longmapsto h^{-1}(a)m$$

is an isomorphism; the target action is $b \cdot m = h^{-1}(b)m$. For $M = s$ it is the canonical simple identification.

In the normalized bar resolution of s , a comparison into its T_h -tensor image applies h^{-1} to every bar letter. It commutes with the differential because h^{-1} preserves products; the first multiplication agrees by the balancing relation for the right-twisted factor, and the last action agrees because h fixes the character of s . The simple-valued degree-three cocycle obtained from p has exactly one dual-ideal input in each nonzero component. Indeed p lowers the dual-letter weight by one, and only weight-zero outputs can act nontrivially on s . The bar comparison therefore multiplies this cochain, and hence its class τ , by H^{-1} .

The tensor functor preserves Yoneda products. In the two-factor algebra A , both factors of one branch use the same H , so both τ_1 and τ_2 acquire the factor H^{-1} . Every basis monomial $\tau_1^r \tau_2^{m-r}$ of H^{3m} acquires H^{-m} . Linearity gives the assertion on the entire homogeneous piece. $\square$

Proposition 8.3. *The finite bimodule F in (8.3) is projective on both A -sides, and the map v of Lemma 8.1 satisfies the following conditions. For every integer $a \geq 0$, define*

$$\delta^a : H^a \oplus H^a \longrightarrow V^a, \quad (g, j) \longmapsto F(g)v + v[a]j. \quad (8.6)$$

Then δ^0 is surjective with a one-dimensional kernel, and δ^a is an isomorphism for every integer $a > 0$.

Proof. Finiteness and both projectivity assertions were proved in the construction. The actual bimodule projections ρ_i are natural with respect to tensoring module maps and their stable shifts. Since $\rho_i v = \text{id}_X$, they give

$$\rho_i[a]F(g)v = U_i(g), \quad \rho_i[a]v[a]j = j.$$

For $a = 0$, k -linearity therefore gives $\delta^0(c \text{id}_X, d \text{id}_X) = (c + d)v$. This is onto and has the diagonal line as its kernel.

Let $a = 3m > 0$. Under (8.5) and Lemma 8.2, the map (8.6) is

$$(g, j) \mapsto (H_1^{-m}g + j, H_2^{-m}g + j). \quad (8.7)$$

It is a 2×2 scalar matrix tensored with the identity of H^{3m} ; that space has dimension $m + 1$. The scalar matrix has determinant $H_1^{-m} + H_2^{-m} \neq 0$. Otherwise $H_1^m = H_2^m$, contrary to algebraic independence: the two distinct monomials remain distinct in the fraction field of $\mathbb{F}_2[q, H_1, H_2]$. This argument holds for every positive integer m , including those divisible by the characteristic. Hence (8.7) is invertible.

If $a > 0$ is not divisible by 3, then $H^a = 0$, and Lemma 8.1 gives $V^a = 0$. Thus δ^a is an isomorphism in these degrees too. $\square$

The hypotheses of Proposition 3.1 are now verified for the finite symmetric algebra A , its finite module X , and the actual finite bimodule F and stable map v . Applying that proposition to

$$\Lambda = \begin{pmatrix} A & 0 \\ F & A \end{pmatrix}$$

produces the required finite nonprojective Gorenstein-projective left module, with both positive Ext vanishings in every degree. In particular, the degree-one assertion uses the surjectivity at degree zero established in Proposition 8.3, in addition to the positive-degree invertibility.

9 The radical and extension of scalars

The construction has already supplied the two Ext vanishings and a finite nonprojective Gorenstein-projective module. We verify the remaining assertions of Theorem 1.1.

Proposition 9.1. *The algebra Λ is noncommutative, satisfies $\Lambda/\text{rad } \Lambda \simeq k^8$, and has $\text{rad}(\Lambda)^4 \neq 0$.*

Proof. Put $J_T = \text{rad } T$. Since $T = C \ltimes DC$ and $C/\text{rad } C \simeq k^2$,

$$J_T = \text{rad } C \oplus DC, \quad T/J_T \simeq k^2.$$

The ideal $J_T \otimes T + T \otimes J_T$ in $A = T \otimes T$ is nilpotent: the two summand ideals commute, and each is nilpotent. Its quotient is k^4 , so it equals $J_A = \text{rad } A$. In the triangular algebra, the ideal

$$J_\Lambda = \begin{pmatrix} J_A & 0 \\ F & J_A \end{pmatrix}$$

is nilpotent as well. Indeed, a product has at most one off-diagonal factor, with powers of J_A on its two sides; sufficiently long products vanish. Its quotient is $k^4 \oplus k^4$. A nilpotent ideal with semisimple quotient is the radical, proving the split assertion and the count of simple modules.

Let $\chi : T \rightarrow k$ be the character of s . There are surjective algebra maps

$$\Lambda \longrightarrow A \xrightarrow{\text{id}_T \otimes \chi} T \longrightarrow C.$$

For a surjection of finite-dimensional algebras, the image of the radical is the radical of the quotient: the image is nilpotent, and the quotient by that image is a quotient of a semisimple algebra. In C , the multiplication table gives

$$(ut)^2 = (y + qx)^2 = q(1 + q)z \neq 0.$$

Since $u, t \in \text{rad } C$, the fourth radical power of C is nonzero, and consequently so is that of Λ . Finally $eu = u$ and $ue = 0$ in C , so C and its covering algebra Λ are noncommutative. $\square$

Proposition 9.2. *For every field extension K/k , the algebra $\Lambda_K = K \otimes_k \Lambda$ and module $Z_K = K \otimes_k Z$ satisfy all conclusions of Theorem 1.1 over K .*

Proof. For finite Λ -modules M, N , take a resolution of M with finite projective terms. In each degree, finite projectivity gives

$$K \otimes_k \text{Hom}_\Lambda(P_i, N) \simeq \text{Hom}_{\Lambda_K}(K \otimes_k P_i, K \otimes_k N).$$

Flatness of K over k commutes with kernels and images, hence

$$K \otimes_k \text{Ext}_\Lambda^i(M, N) \simeq \text{Ext}_{\Lambda_K}^i(M_K, N_K) \quad (i \geq 0). \quad (9.1)$$

This proves both Ext vanishings. The totally acyclic complex defining Z remains exact after tensoring with K ; the same termwise Hom isomorphism, now with target Λ , shows that its dual complex over Λ_K is exact. Thus Z_K is Gorenstein-projective.

Choose a finite projective surjection $P \rightarrow Z$ with kernel N . It does not split, because Z is nonprojective. Its class in $\text{Ext}_\Lambda^1(Z, N)$ is nonzero and stays nonzero under (9.1), since field extension is faithful. Thus the extended sequence does not split and Z_K is nonprojective.

The radical ideal in Proposition 9.1 stays nilpotent and has quotient K^8 after scalar extension, so it is the radical of Λ_K . Its nonzero fourth power stays nonzero. This also proves the split and radical assertions over K . $\square$

A The complete degree-three cochain

This appendix specifies the finite cochain used in Lemma 4.4, including the coefficient identities that verify its properties. Lower-case letters are the basis of C from (4.1); capital letters are the dual basis of DC . A four-letter entry $abcv$ in the table means that the indicated coefficient is the coefficient of v in $p(a, b, c)$. All unlisted coefficients are zero. For example, the entry tyJf in the q^3 row means $p(t, y, J) = q^3 f$. Each entry has composable radical inputs and an output with the same two endpoints.

coefficient	input and output letters									
q^2	EuNT	EyZX	FNtU	JtXZ	NtEU	NtXV	XzJT	ZxEY	tEun	tXuf
q^2	tXvn	tXzj	tuFn	tuNf	tvNn	uFNT	vNNT	vtJu	xEyz	zJtx
q	EuUE	EuVX	EvVE	ExXE	EyYE	EzZE	FNnF	FVvF	FVxU	FtTF
q	FtYU	Ftun	Ftxj	JFtX	JNtZ	JUyX	JVyZ	JjEE	JjTT	JjXX
q	NFtU	NNtV	NUyU	NVyV	NjJN	NjZV	NnNN	NnVV	Ntuf	Ntvn
q	Ntzj	TtEE	TtXX	UuFF	UuUU	UvNF	UvVU	UxXU	Uyun	Uyxj
q	UzZU	VvNN	VvVV	VxEU	VxXV	Vyuf	Vyvn	Vyzj	VzZV	XxEE
q	XxXX	YuFT	YuUY	YvNT	YvVY	YxXY	YyEE	YyJJ	YyTT	YyXX
q	YyYY	YyZZ	YzZY	ZxXZ	ZzEE	ZzJJ	ZzTT	ZzXX	jJFF	jJUJ
q	jJjj	jJnn	jYyj	jZvn	jZzj	nNFF	nNUU	ntTn	tJjt	tTNN
q	tTVV	tTtt	tYyt	tZzt	tzJn	uFVX	uFty	uNFT	uNUY	uNte
q	uUyy	uVye	ujJu	vNUE	vNVX	vNty	vVyy	vtTv	xEuv	xJFT
q	xJUy	xJjx	xJte	xTtx	xXuu	xXvv	xXyy	xXzz	xYyx	xZye
q	xZzx	xuFv	xuNu	xvNv	yxJu	yzJv	zJNT	zJvY	zYyz	zZuu
q	zZvv	zZyy	zZzz							
q^3	EyJT	JtEY	tEyj	tyJf						
1	JNjX	JVzX	NNjU	NVzU	NjYU	Njun	Njxj	Ntxt	TNtX	TVyX
1	Vyxt	VzYU	Vzun	Vzxj	YuNJ	YuVZ	jYun	juNn	jxJn	txTn
1	uNNJ	uNVZ	uNjy	uVzy	xJNJ	xJVZ	xTNT	xTVY	yxTv	zYuv
1	zuNv	zxJv								

There are 179 entries, with distinct three-letter inputs. In each entry the number of capital input letters is one more than the number of capital output letters. This proves the weight assertion.

Here are explicit finite recurrences for checking the remaining identities. They also specify how every zero entry participates in the calculation. Let $\mathcal{B}$ be the twenty-letter basis and $\mathcal{B}_\tau = \mathcal{B} \setminus \{e, f\}$. Write

$$ab = \sum_v \mu_{ab}^v v, \quad p(a, b, c) = \sum_v p_{abc}^v v.$$

The lower-case coefficients μ are exactly (4.2) and the idempotent actions. The remaining coefficients are determined, for lower-case a, b, c , by

$$\mu_{a,b^*}^{c^*} = \mu_{ca}^b, \quad \mu_{b^*,a}^{c^*} = \mu_{ac}^b, \quad \mu_{a^*,b^*}^v = 0. \quad (\text{A.1})$$

Unspecified coefficients are zero. In particular every coefficient belongs to $\mathbb{F}_2[q]$.

For each $a, b, c, d \in \mathcal{B}_\tau$ and $v \in \mathcal{B}$, substitution of the table and (A.1) gives

$$0 = \sum_{x \in \mathcal{B}} (\mu_{ax}^v p_{bcd}^x + \mu_{ab}^x p_{xcd}^v + \mu_{bc}^x p_{axd}^v + \mu_{cd}^x p_{abx}^v + p_{abc}^x \mu_{xd}^v). \quad (\text{A.2})$$

This is the coefficient of v in the normalized Hochschild differential. One may check it on all $18^4 = 104\,976$ radical four-words; precisely 15 250 have composable endpoints, and the other equations vanish by the corners. Thus the finite calculation covers the whole relative bar term, not a selected collection of words.

For completeness, the boundary identity has an equally short coefficient check. Set $\epsilon_a = 0$ on lower-case letters and $\epsilon_a = 1$ on capital letters. For every basis letter put

$$\gamma_a = \begin{cases} q, & a \in \{u, v, t, E, X, Y, Z, U, V\}, \\ 0, & \text{otherwise.} \end{cases}$$

For all $a, b \in \mathcal{B}_\tau$ and $v \in \mathcal{B}$, the same table gives

$$\sum_{\substack{w \in \mathcal{B}_\tau \\ \epsilon_w=0}} \sum_x p_{abw}^x \mu_{x,w^*}^v = \epsilon_b \gamma_a \mu_{ab}^v, \quad (\text{A.3})$$

$$\sum_{\substack{w \in \mathcal{B}_\tau \\ \epsilon_w=1}} \sum_x p_{abw}^x \mu_{x,w^*}^v = (\gamma_b + \gamma_v + (1 - \epsilon_b) \gamma_a) \mu_{ab}^v. \quad (\text{A.4})$$

There are $18^2 = 324$ input pairs. These two equations are the constant and linear coefficients in H of (4.6): its left side gives H^{ϵ_w} , while the three terms on the right give respectively $H\gamma_b\mu_{ab}^v$, $H\gamma_v\mu_{ab}^v$, and $H^{1-\epsilon_b}\gamma_a\mu_{ab}^v$. This verifies the identity for an indeterminate H , hence for either H_1 or H_2 .

All these substitutions are polynomial identities in characteristic two. They require no specialization of q or H . Finally the entries $p(t, x, J) = 0$ and $p(t, y, J) = q^3 f$ give

$$q^2 p(t, x, J) + p(t, y, J) = q^3 f,$$

which is the nonzero value on the explicit cycle (4.7).

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